

CIS6930 - Project Proposal

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Research Question

Research Question: Does schema-constrained validation with retry loops reduce the output volatility of a GPT-4-orchestrated multi-source integration pipeline by at least 50% compared to unconstrained generation, measured by row-level Jaccard similarity across repeated runs on U.S. travel advisory data for 30 countries?

Relevance to Data Engineering: LLM-orchestrated pipelines offer flexibility for extraction and schema mapping but introduce volatility — identical inputs can yield different risk ratings, normalizations, or hallucinated fields across runs. If architectural guardrails can demonstrably stabilize these outputs, it opens a practical path for adopting LLM pipelines in production as a governed complement to deterministic logic.

Project Direction

Direction: C — Novel Architecture with Stability Focus. The pipeline wraps an LLM orchestrator in layered stability constraints (schema validation, retry loops, entity normalization) and measures their effect on output volatility under repeated execution and input perturbation.

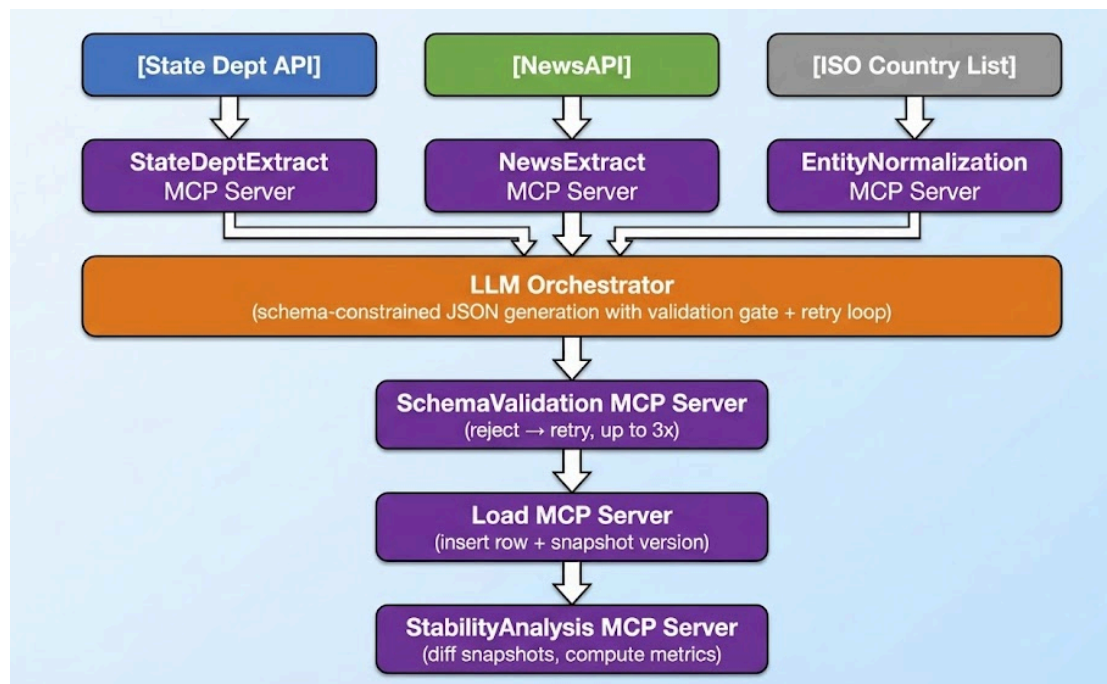
Dataset. Three sources produce structured, country-level travel risk assessments:

1. U.S. State Department Travel Advisories with official risk levels 1–4 for 30 countries.
2. Five NewsAPI articles per country, introducing realistic input noise.
3. ISO 3166 country metadata for entity normalization. Articles are fetched once and cached locally to ensure reproducibility.

The 30-country scope keeps experimental runs affordable while spanning all four State Department risk tiers. Articles are fetched once and cached locally to ensure reproducibility regardless of API availability.

Initial Design

Architecture Diagram



MCP Servers. Five servers separate concerns. **StateDeptExtractServer** and **NewsExtractServer** perform deterministic extraction from advisories and news, respectively. **SchemaValidationServer** is the core constraint under study: it validates LLM-generated JSON against a strict schema (required fields, value constraints, hallucinated field rejection) and triggers up to three retries on failure. **LoadServer** inserts validated rows into SQLite with versioned snapshots. **StabilityAnalysisServer** diffs snapshots across runs and computes the metrics below.

LLM Role. GPT-4 receives structured extractions from the deterministic servers and synthesizes them into unified risk assessments. It decides how to weight advisory vs. news signals and generates schema-constrained JSON. The LLM never writes directly to the database; every row must pass validation first.

Evaluation Plan

Stability (Primary). *Reproducibility rate:* three identical runs, row-level Jaccard similarity field-by-field. *Schema compliance:* first-attempt pass rate and average retries. *Entity stability:* add/remove one article per country, track risk_level changes. *Sensitivity:* risk score variance across five perturbed runs; summary embedding similarity (all-MiniLM-L6-v2) with a 0.90 threshold.

Accuracy (Secondary). Precision/recall for risk_level, event_types, and citation validity on a 10-country ground truth subset. Two annotators label independently; agreement measured with Cohen's kappa.

Cost. Token usage, API calls, runtime, and dollar cost per run — reported per-country and pipeline-wide.

Baselines. (1) *Deterministic ETL:* regex parsing, keyword classification, fuzzy matching, template summaries — the stability ceiling. (2) *Single-shot LLM (stretch):* one prompt, no validation — isolates the effect of constraints.

Timeline

Week	Milestone	Key Tasks
1	Proposal Finalization	Finalize schema, risk taxonomy, and perturbation strategy.
2	Design Review & Data Acquisition	Collect and cache data for 30 countries. Finalize system architecture.
3–4	Code Checkpoint & Baseline Implementation	Build all Multi-Cluster Processor (MCP) servers, implement orchestrator with retry loop, integrate SQLite snapshots, and establish the deterministic baseline. Execute the first end-to-end run.
5	Draft Paper Completion	Conduct perturbation experiments, perform ground truth annotation, compute all required metrics, and complete the draft paper.
6–7	Final Paper Submission	Perform comparative analysis, implement a single-shot baseline (if time permits), and polish the final paper.
8	Final Presentation	Deliver a 10-minute presentation followed by a Q&A session.